

Warm-Up Quiz — Day 10

Inequalities, Limits, and Convergence in Probability

Complete before lecture. 10 minutes.

Name: _____

Today we study the foundations of large-sample theory: Markov and Chebyshev inequalities, the law of large numbers, and convergence in probability. This quiz reviews the probability tools behind those results.

1. Markov's Inequality.

If $X \geq 0$ and $\mathbb{E}[X] = 4$, Markov's inequality says $\Pr(X \geq a) \leq \mathbb{E}[X]/a$.

- (a) What upper bound does Markov give for $\Pr(X \geq 20)$?
- (b) Can you conclude anything about $\Pr(X \geq 2)$? Is the bound useful here?

2. Chebyshev's Inequality.

If $\mathbb{E}[X] = \mu$ and $\text{Var}(X) = \sigma^2$, then $\Pr(|X - \mu| \geq k) \leq \sigma^2/k^2$.

- (a) Let $\mu = 10$, $\sigma^2 = 4$. Bound $\Pr(|X - 10| \geq 6)$.
- (b) Now let $\bar{X}_n$ be the sample mean of n iid draws from this distribution. What is $\text{Var}(\bar{X}_n)$?
- (c) Apply Chebyshev to $\bar{X}_n$: bound $\Pr(|\bar{X}_n - 10| \geq 0.1)$ for $n = 10,000$.

3. Convergence in Probability.

We say $X_n \xrightarrow{P} c$ if $\Pr(|X_n - c| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$.

- (a) Using your answer from 2(c), argue that $\bar{X}_n \xrightarrow{P} \mu$ as $n \rightarrow \infty$. (This is the WLLN.)

- (b) If $X_n \xrightarrow{p} 5$ and $Y_n \xrightarrow{p} 2$, what does X_n/Y_n converge to in probability? (Name the theorem.)

4. Variance of a Sum (CLT Preview).

Let $X_1, \dots, X_n$ be iid with $\mathbb{E}[X_i] = 0$ and $\text{Var}(X_i) = \sigma^2$.

- (a) Compute $\text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right)$.
- (b) Compute $\text{Var}\left(\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i\right)$. Why doesn't this vanish as $n \rightarrow \infty$?

Answer Key — Day 10

1. (a) $\Pr(X \geq 20) \leq 4/20 = 0.2$.
(b) $\Pr(X \geq 2) \leq 4/2 = 2$. Since probabilities are at most 1, this bound is vacuous—it tells us nothing.
2. (a) $\Pr(|X - 10| \geq 6) \leq 4/36 = 1/9 \approx 0.111$.
(b) $\text{Var}(\bar{X}_n) = \sigma^2/n = 4/n$.
(c) $\Pr(|\bar{X}_n - 10| \geq 0.1) \leq \frac{4/10000}{0.01} = \frac{0.0004}{0.01} = 0.04$.
3. (a) By Chebyshev, $\Pr(|\bar{X}_n - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2) \rightarrow 0$ as $n \rightarrow \infty$ for any fixed $\varepsilon > 0$. This is exactly the definition of $\bar{X}_n \xrightarrow{P} \mu$.
(b) $X_n/Y_n \xrightarrow{P} 5/2$. This follows from Slutsky's theorem (or the continuous mapping theorem applied to $g(x, y) = x/y$, which is continuous at $(5, 2)$).
4. (a) $\text{Var}(\frac{1}{n} \sum X_i) = \frac{1}{n^2} \cdot n\sigma^2 = \sigma^2/n$.
(b) $\text{Var}(\frac{1}{\sqrt{n}} \sum X_i) = \frac{1}{n} \cdot n\sigma^2 = \sigma^2$. The $\sqrt{n}$ normalization exactly balances the accumulating variance, keeping it constant. This is why the CLT uses $\sqrt{n}$: it is the unique scaling that produces a non-degenerate limit.